1 Confidence Intervals, etc.

If a value z_c or t_c value with $N-1$ degrees of freedom corresponds to our desired confidence interval:

$$\mu \in \left(\bar{X} - \frac{z_c \sigma}{\sqrt{N}}, \bar{X} + \frac{z_c \sigma}{\sqrt{N}} \right)$$

If it is a one tailed test, we use $\alpha = 1 - (confidence_{interval})$, but for a two-tailed test, we use $\alpha = \frac{\alpha_{onetail}}{2}$

2 RegEx

In other document.

3 Linear Regression

$$\beta = Vy$$

Where

$$V = (X^T X)^{-1} X^T$$

Assume $\beta = [a_1, a_2, \ldots, b]^T$.

$$\hat{y} = a_1 x_1 + a_2 x_2 + \cdots + b$$

If there are N samples, with M features + bias term (b).
Without bias term:

$$\begin{aligned} \dim(X) &= N \times M \\ \dim(y) &= N \times 1 \\ \dim(\beta) &= M \times 1 \end{aligned}$$

That is X has N rows, M columns.
With bias term:

$$\begin{aligned} \dim(X) &= N \times (M+1) \\ \dim(y) &= N \times 1 \\ \dim(\beta) &= (M+1) \times 1 \end{aligned}$$

Essentially, the bias term is an additional feature when looking at dimensions.
If there is no bias term initially, or provided, one can assume all bias terms have a value of 1, making it an additional last column to the X matrix.

3.1 Normalization

We calculate μ and σ for each column in X and y where

$$\begin{aligned} \mu_n &= \frac{\sum x}{\text{len}(x_n)} \\ \sigma_n &= \sqrt{\frac{\sum (x-\mu)^2}{\text{len}(x_n)}} \end{aligned}$$

Normalize using:

$$z = \frac{x - \mu}{\sigma}$$

3.2 MSE

$$MSE = \frac{\sum E^2}{N}$$

where E is the error of a sample, and N is the number of samples.

4 Ridge Regression

$$\beta = Vy$$

Where

$$V = (X^T X + \lambda I)^{-1} X^T$$

If we want to perform N-fold cross-validation, and we are considering M values of λ, there are MN possibilities of trainable models.
The larger the value of λ, the less it fits the model. So, a very small lambda is likely to overfit, or, at the very least, allow for overfitting.
Ridge Regression Loss is:

$$\sum (y - \hat{y})^2 + \lambda \sum \beta^2$$

5 N-gram

- word-based: N consecutive words, starting from one word after the previous one. Will have exactly N words.
- character-based: N consecutive characters, rest is basically the same as the above.

Term frequency:

$$TF(t, d) = \frac{\#t \text{ in } d}{\text{len}(d)}$$

Inverse Document Frequency:

$$IDF(t, D) = \log\left(\frac{\#\text{documents}}{\#\text{documents with } t}\right)$$

Importance of word:

$$TF\text{-}IDF(t, d, D) = TF(t, d) \times IDF(t, D)$$